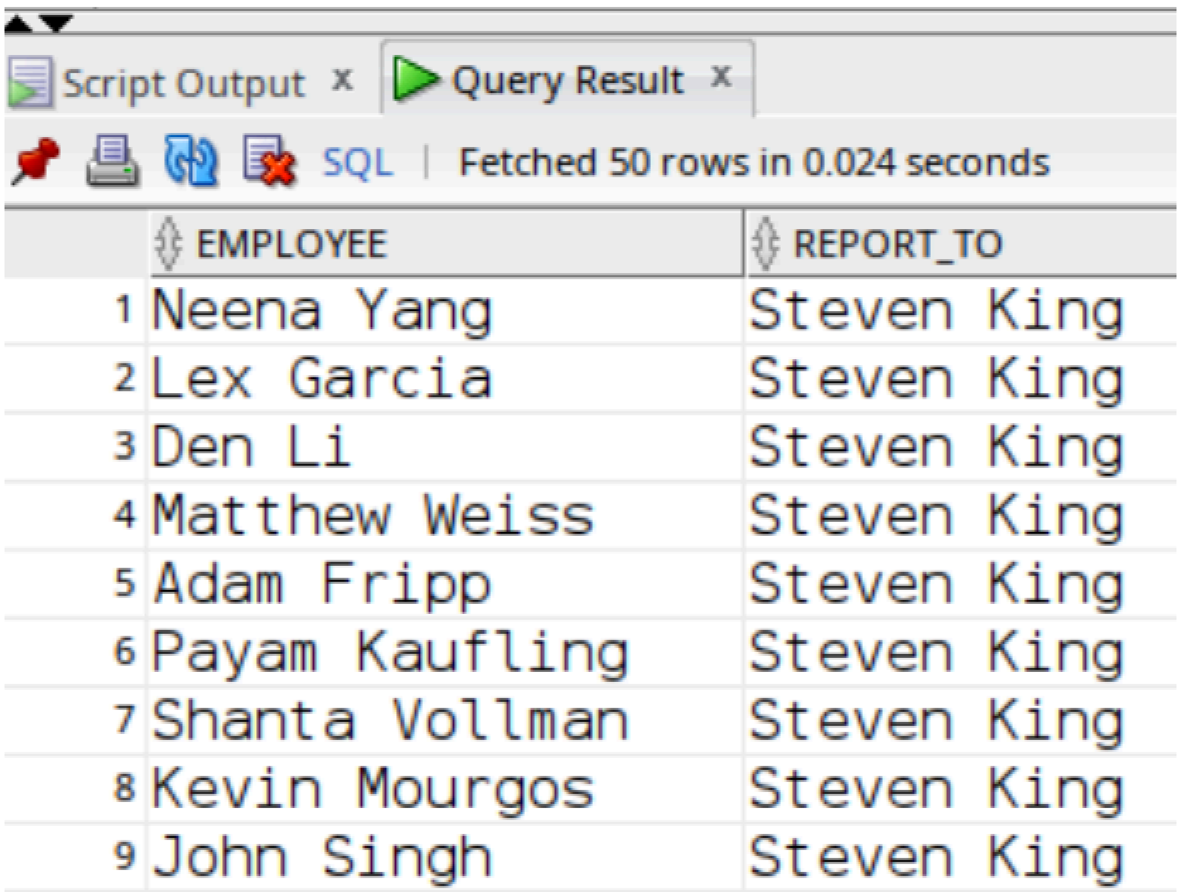
Q1:

select e.first_name || ' ' || e.last_name as employee,
m.first_name || ' ' || m.last_name as report_to
from employees e
left join employees m on e.manager_id=m.employee_id;



	EMPLOYEE	REPORT_TO
1	Neena Yang	Steven King
2	Lex Garcia	Steven King
3	Den Li	Steven King
4	Matthew Weiss	Steven King
5	Adam Fripp	Steven King
6	Payam Kaufling	Steven King
7	Shanta Vollman	Steven King
8	Kevin Mourgos	Steven King
9	John Singh	Steven King

Q2:

select e.first_name,e.last_name,e.salary,m.first_name,m.last_name,m.salary
from employees e join employees m on e.manager_id=m.employee_id
where e.salary>m.salary;

Script Output × | Query Result ×

SQL | All Rows Fetched: 2 in 0.009 seconds

	FIRST_NAME	LAST_NAME	SALARY	FIRST_NAME_1	LAST_NAME_1	SALARY_1
1	Lisa	Ozer	11500	Gerald	Cambrault	11000
2	Ellen	Abel	11000	Eleni	Zlotkey	10500

Q3:

select department_id,department_name from departments where department_id not in (select department_id from employees where department_id is not null);

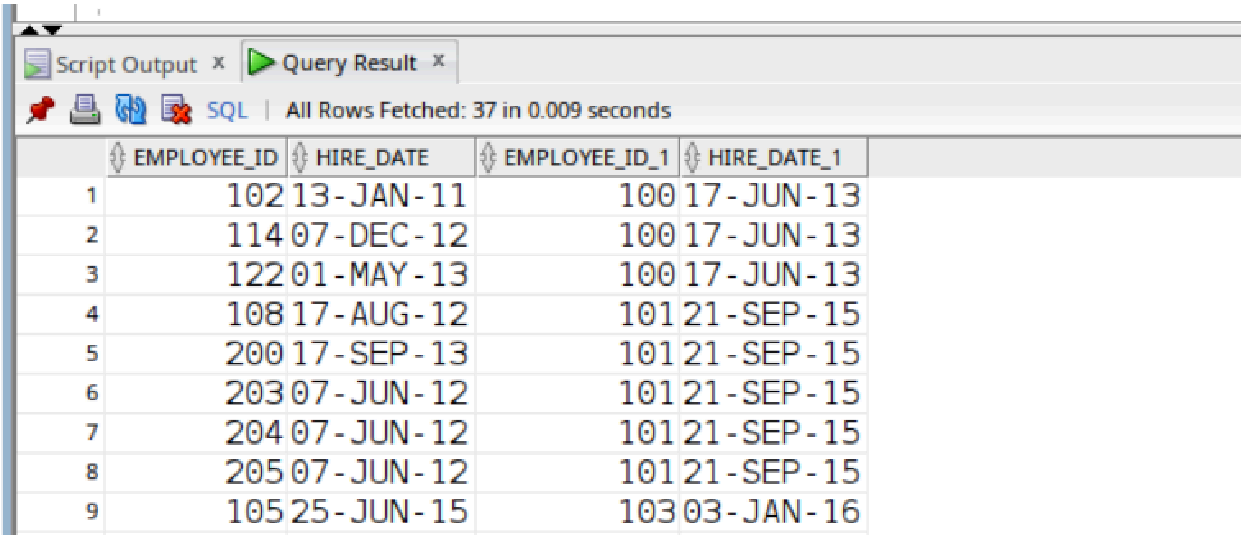
Script Output × | Query Result ×

SQL | All Rows Fetched: 16 in 0.04 seconds

	DEPARTMENT_ID	DEPARTMENT_NAME
1	120	Treasury
2	130	Corporate Tax
3	140	Control And Credit
4	150	Shareholder Services
5	160	Benefits
6	170	Manufacturing
7	180	Construction
8	190	Contracting
9	200	Operations

Q4:

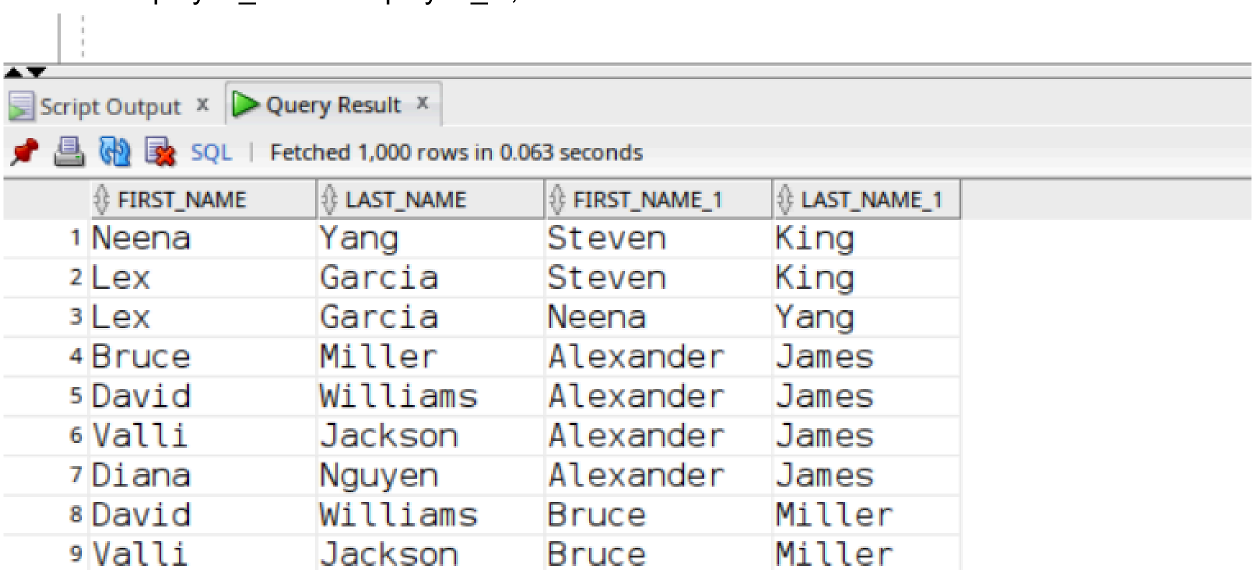
select e.employee_id,e.hire_date,m.employee_id,m.hire_date from employees e
join employees m on e.manager_id=m.employee_id where e.hire_date<m.hire_date;



	EMPLOYEE_ID	HIRE_DATE	EMPLOYEE_ID_1	HIRE_DATE_1
1	102	13-JAN-11	100	17-JUN-13
2	114	07-DEC-12	100	17-JUN-13
3	122	01-MAY-13	100	17-JUN-13
4	108	17-AUG-12	101	21-SEP-15
5	200	17-SEP-13	101	21-SEP-15
6	203	07-JUN-12	101	21-SEP-15
7	204	07-JUN-12	101	21-SEP-15
8	205	07-JUN-12	101	21-SEP-15
9	105	25-JUN-15	103	03-JAN-16

Q5:

select e.first_name,e.last_name,m.first_name,m.last_name from employees e
join employees m on e.department_id=m.department_id
where e.employee_id > m.employee_id;



	FIRST_NAME	LAST_NAME	FIRST_NAME_1	LAST_NAME_1
1	Neena	Yang	Steven	King
2	Lex	Garcia	Steven	King
3	Lex	Garcia	Neena	Yang
4	Bruce	Miller	Alexander	James
5	David	Williams	Alexander	James
6	Valli	Jackson	Alexander	James
7	Diana	Nguyen	Alexander	James
8	David	Williams	Bruce	Miller
9	Valli	Jackson	Bruce	Miller

Q6:

select e.*,l.city from employees e
join departments d on e.department_id=d.department_id
join locations l on d.location_id=l.location_id;

Script Output × Query Result ×

SQL | Fetched 50 rows in 0.016 seconds

	EMPLOYEE_ID	FIRST_NAME	LAST_NAME	EMAIL	PHONE_NUMBER	HIRE_DATE	JOB_ID	SALARY	COMMISSION_PCT	MANAGER_ID	DEPARTMENT_ID	CITY
1	100	Steven	King	SKING	1.515.555.0100	17-JUN-13	AD_PRES	24000	(null)	(null)	90	Seattle
2	101	Neena	Yang	NYANG	1.515.555.0101	21-SEP-15	AD_VP	17000	(null)	100	90	Seattle
3	102	Lex	Garcia	LGARCIA	1.515.555.0102	13-JAN-11	AD_VP	17000	(null)	100	90	Seattle
4	103	Alexander	James	AJAMES	1.590.555.0103	03-JAN-16	IT_PROG	9000	(null)	102	60	Southlake
5	104	Bruce	Miller	BMILLER	1.590.555.0104	21-MAY-17	IT_PROG	6000	(null)	103	60	Southlake
6	105	David	Williams	DWILLIAMS	1.590.555.0105	25-JUN-15	IT_PROG	4800	(null)	103	60	Southlake
7	106	Valli	Jackson	VJACKSON	1.590.555.0106	05-FEB-16	IT_PROG	4800	(null)	103	60	Southlake
8	107	Diana	Nguyen	DNGUYEN	1.590.555.0107	07-FEB-17	IT_PROG	4200	(null)	103	60	Southlake
9	108	Nancy	Gruenberg	NGRUENBE	1.515.555.0108	17-AUG-12	FI_MGR	12008	(null)	101	100	Seattle

Q7:

select e.*,m.department_id from employees e join employees m
on e.department_id=m.department_id where m.first_name='steven'
and m.last_name='king' and e.employee_id <> m.employee_id;

Script Output × Query Result ×

SQL | All Rows Fetched: 2 in 0.009 seconds

	EMPLOYEE_ID	FIRST_NAME	LAST_NAME	EMAIL	PHONE_NUMBER	HIRE_DATE	JOB_ID	SALARY	COMMISSION_PCT	MANAGER_ID	DEPARTMENT_ID	DEPARTMENT_ID_1
1	101	Neena	Yang	NYANG	1.515.555.0101	21-SEP-15	AD_VP	17000	(null)	100	90	90
2	102	Lex	Garcia	LGARCIA	1.515.555.0102	13-JAN-11	AD_VP	17000	(null)	100	90	90

Q8:

select e.* from employees e left join
(select department_id, avg(salary) as avg_salary from employees group by department_id) m
on e.department_id=m.department_id
where e.salary > m.avg_salary;

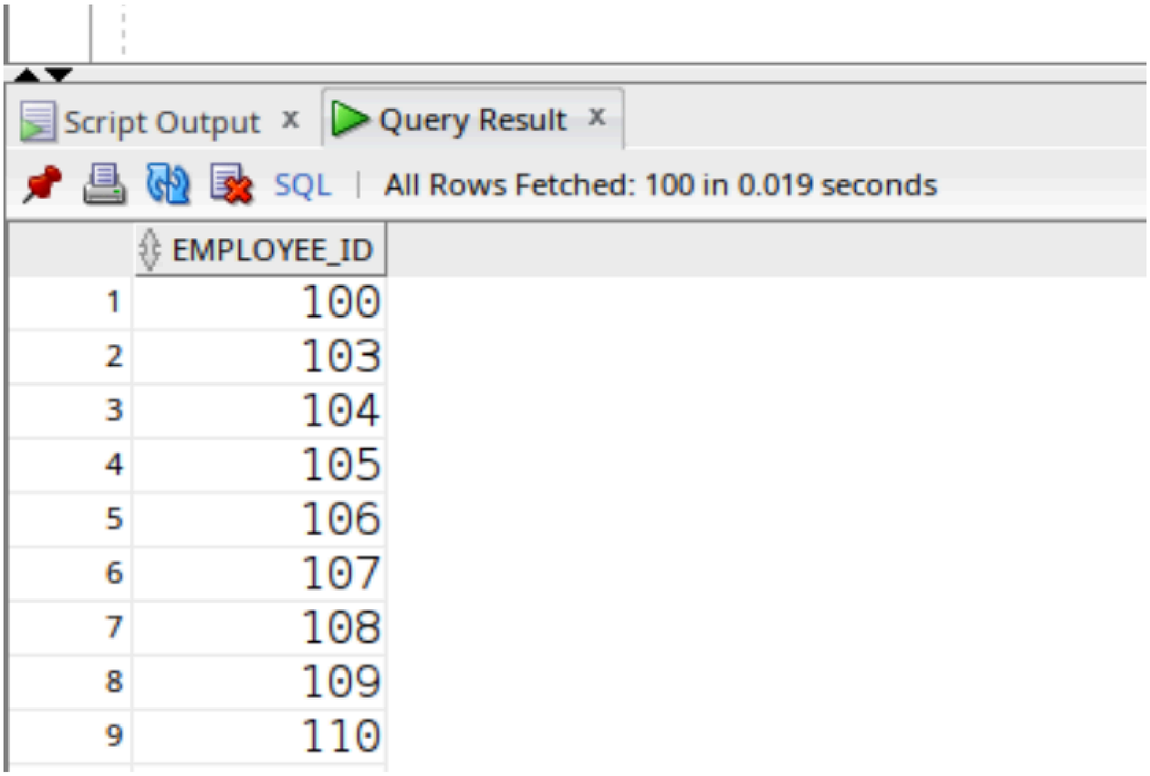
cript Output × Query Result ×

SQL | All Rows Fetched: 38 in 0.059 seconds

	EMPLOYEE_ID	FIRST_NAME	LAST_NAME	EMAIL	PHONE_NUMBER	HIRE_DATE	JOB_ID	SALARY	COMMISSION_PCT	MANAGER_ID	DEPARTMENT_ID
1	100	Steven	King	SKING	1.515.555.0100	17-JUN-13	AD_PRES	24000	(null)	(null)	90
2	103	Alexander	James	AJAMES	1.590.555.0103	03-JAN-16	IT_PROG	9000	(null)	102	60
3	104	Bruce	Miller	BMILLER	1.590.555.0104	21-MAY-17	IT_PROG	6000	(null)	103	60
4	108	Nancy	Gruenberg	NGRUENBE	1.515.555.0108	17-AUG-12	FI_MGR	12008	(null)	101	100
5	109	Daniel	Faviet	DFAVIET	1.515.555.0109	16-AUG-12	FI_ACCOUNT	9000	(null)	108	100

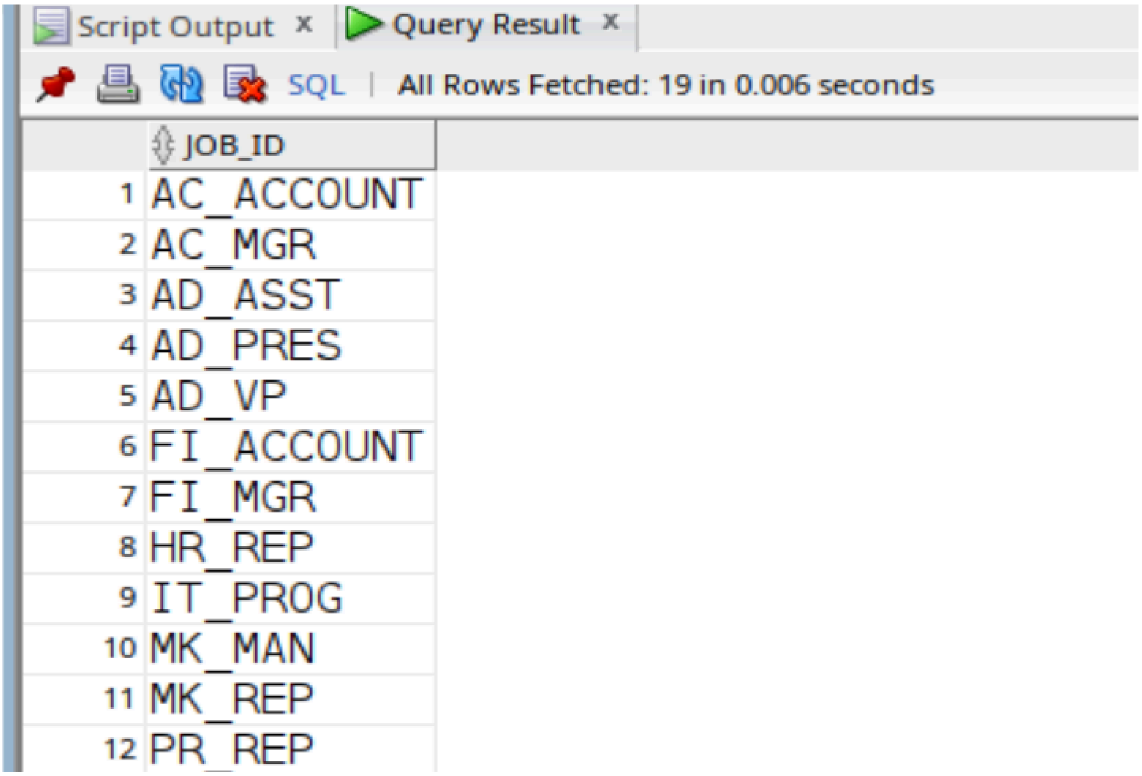
Q9:

select employee_id from employees
minus
select employee_id from job_history;



Q10:

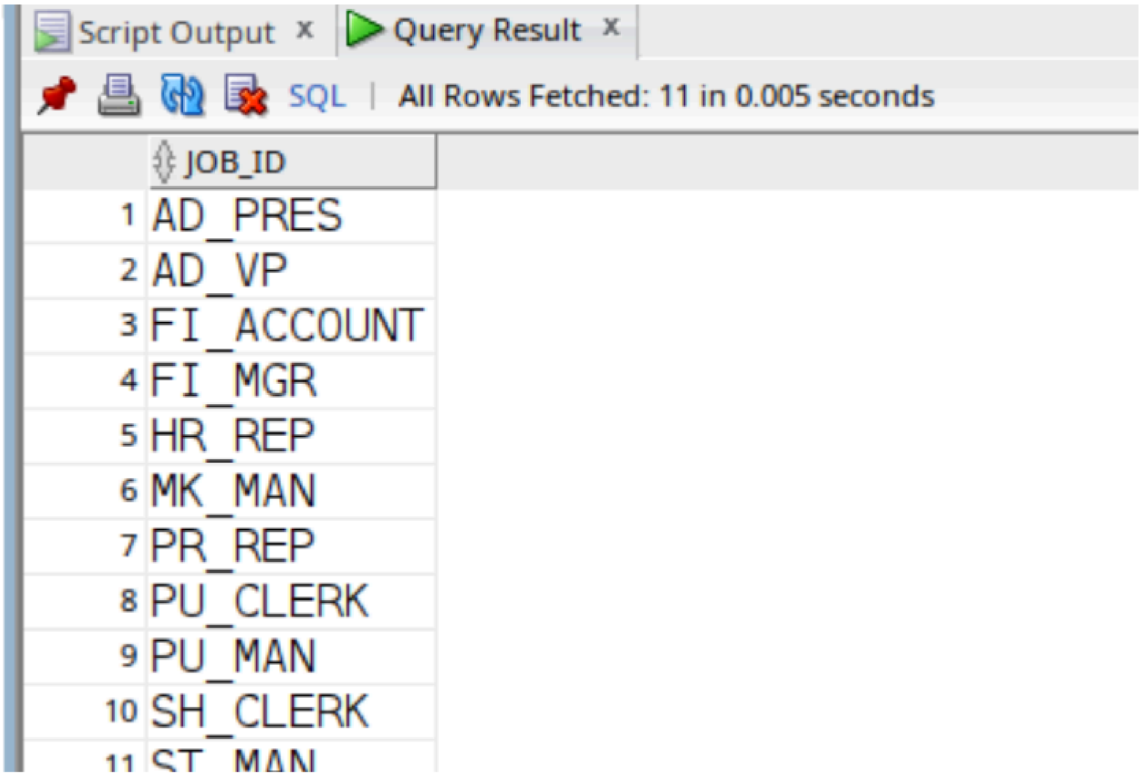
select job_id from employees
union
select job_id from job_history;



	JOB_ID
1	AC_ACCOUNT
2	AC_MGR
3	AD_ASST
4	AD_PRES
5	AD_VP
6	FI_ACCOUNT
7	FI_MGR
8	HR_REP
9	IT_PROG
10	MK_MAN
11	MK_REP
12	PR_REP

Q11:

```
select job_id from employees
minus
select job_id from job_history;
```



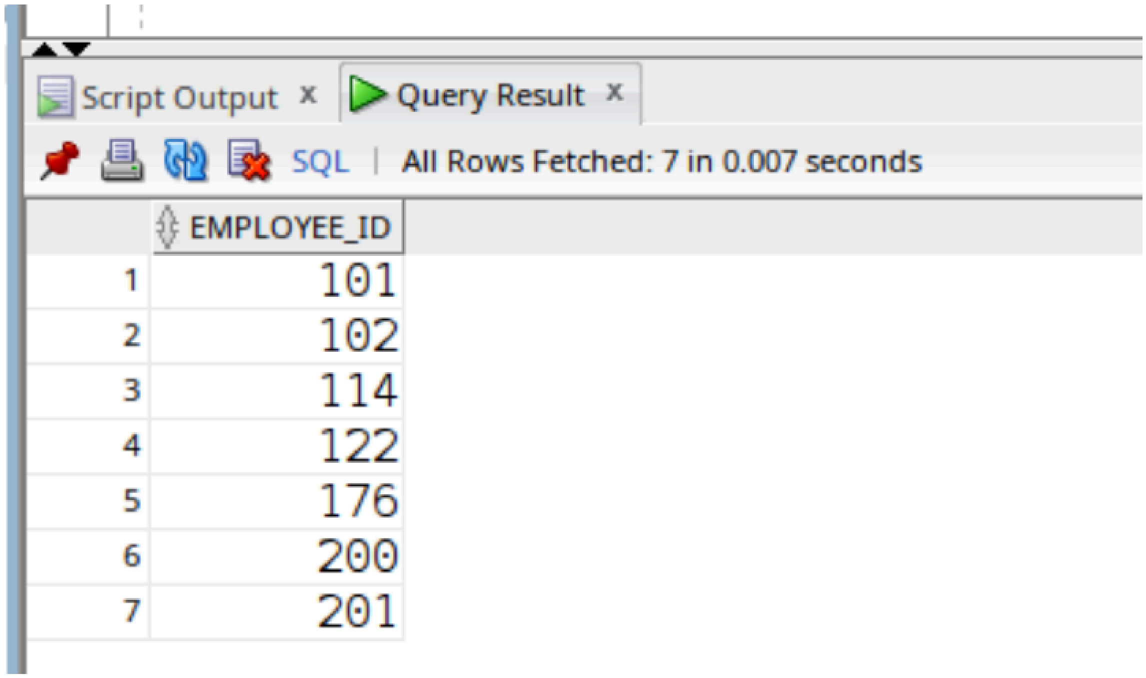
Script Output × | Query Result ×

SQL | All Rows Fetched: 11 in 0.005 seconds

	JOB_ID
1	AD_PRES
2	AD_VP
3	FI_ACCOUNT
4	FI_MGR
5	HR_REP
6	MK_MAN
7	PR_REP
8	PU_CLERK
9	PU_MAN
10	SH_CLERK
11	ST_MAN

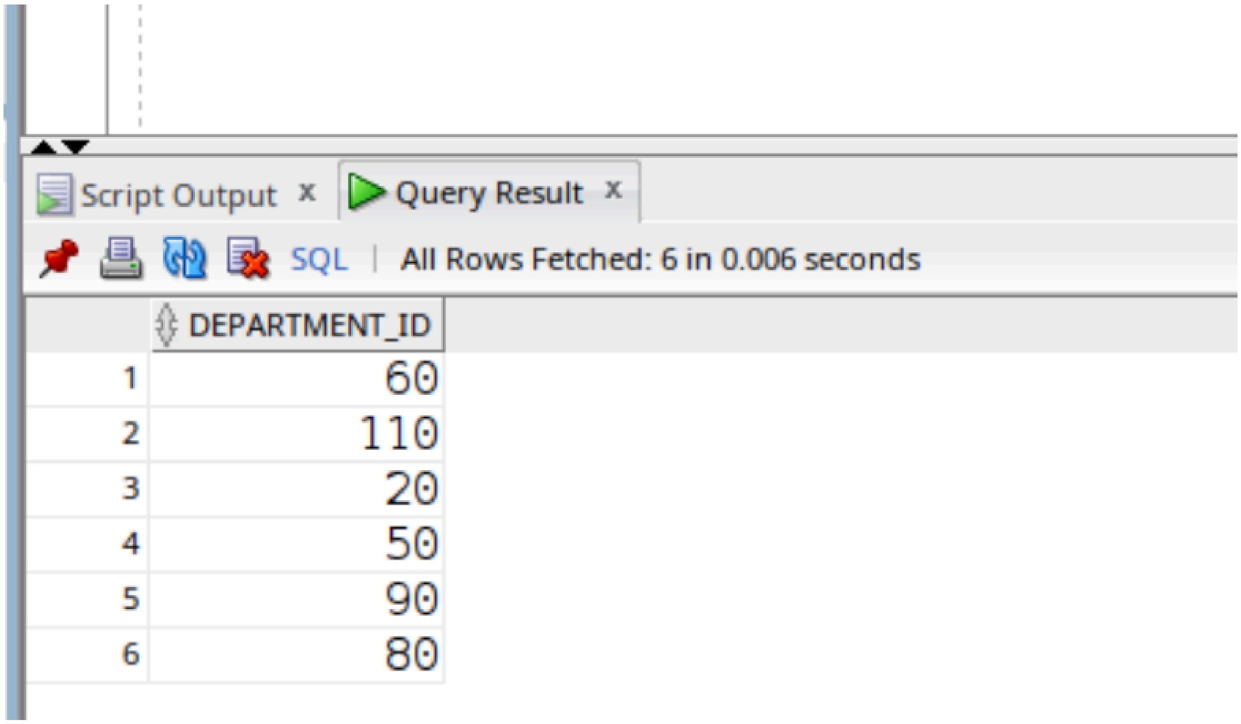
Q12:

```
select employee_id from employees
intersect
select employee_id from job_history;
```



Q13:

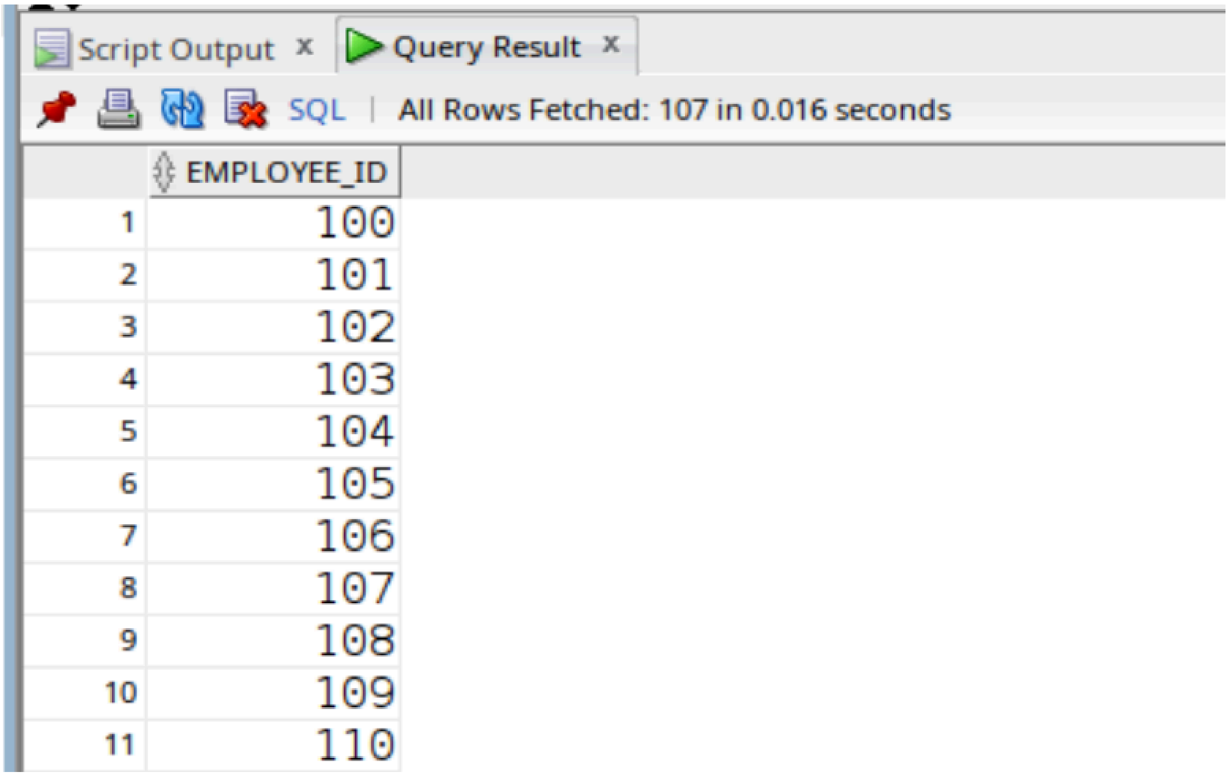
select department_id from employees
intersect
select department_id from job_history;



	DEPARTMENT_ID
1	60
2	110
3	20
4	50
5	90
6	80

Q14:

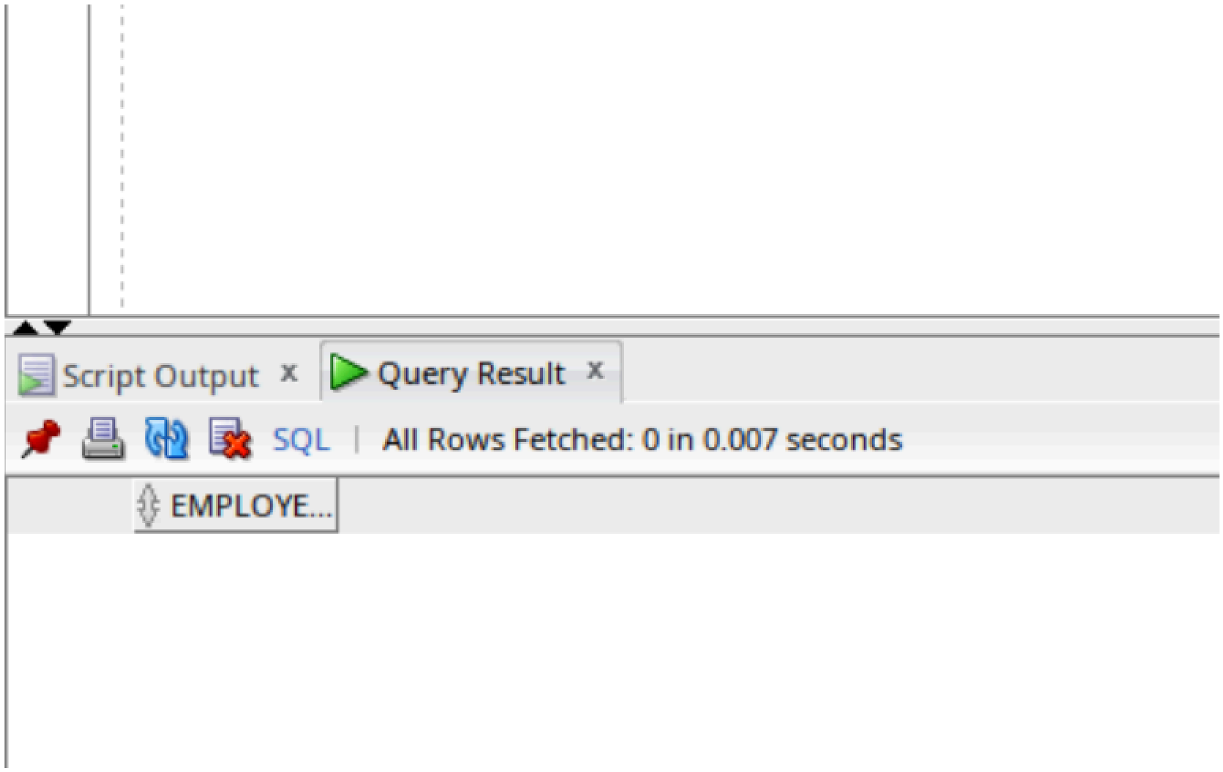
```
select employee_id from employees
union
select employee_id from job_history;
```



	EMPLOYEE_ID
1	100
2	101
3	102
4	103
5	104
6	105
7	106
8	107
9	108
10	109
11	110

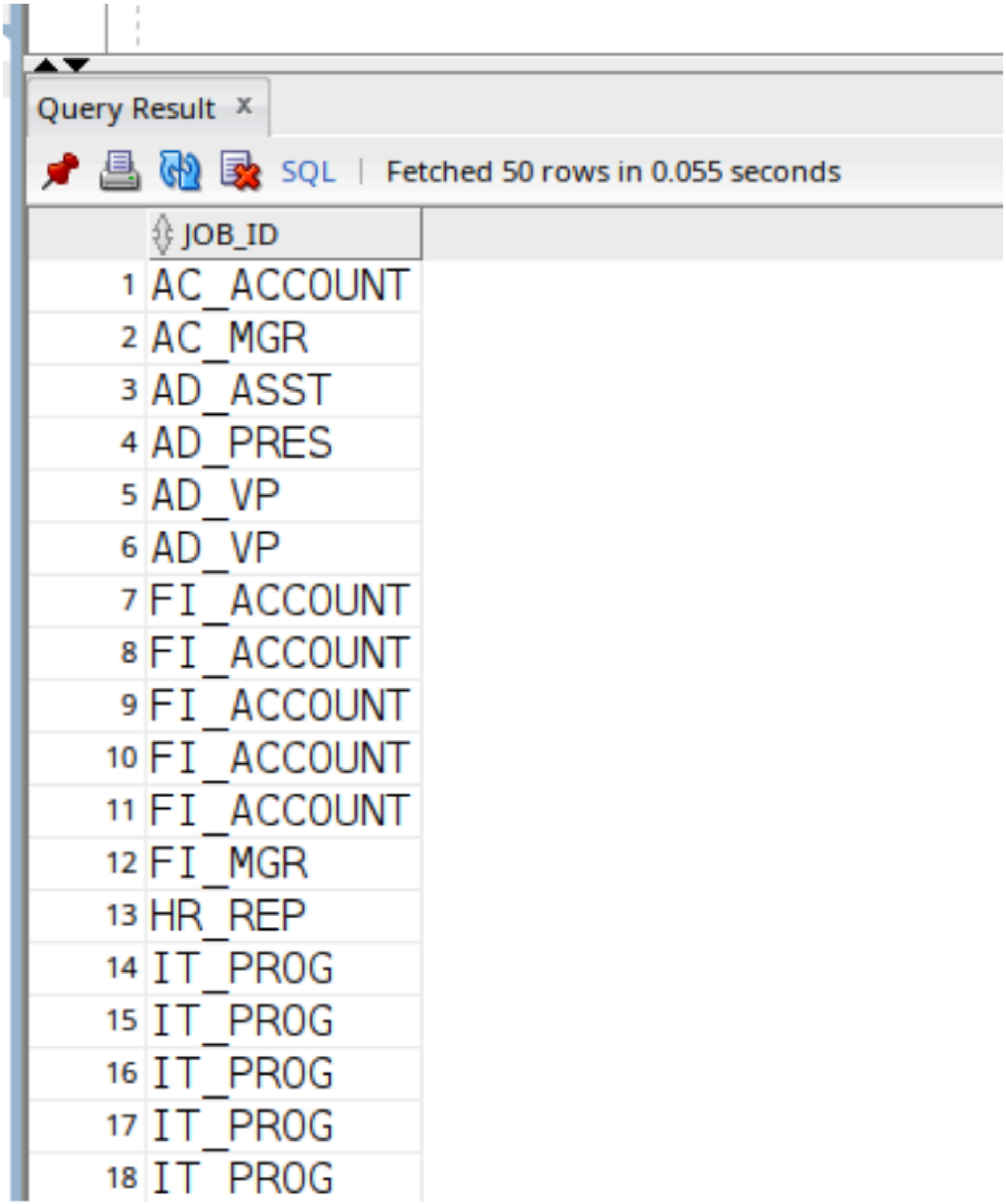
Q15:

```
select employee_id from job_history
minus
select employee_id from employees;
```



Q16:

```
select job_id from employees
union all
select job_id from job_history;
```



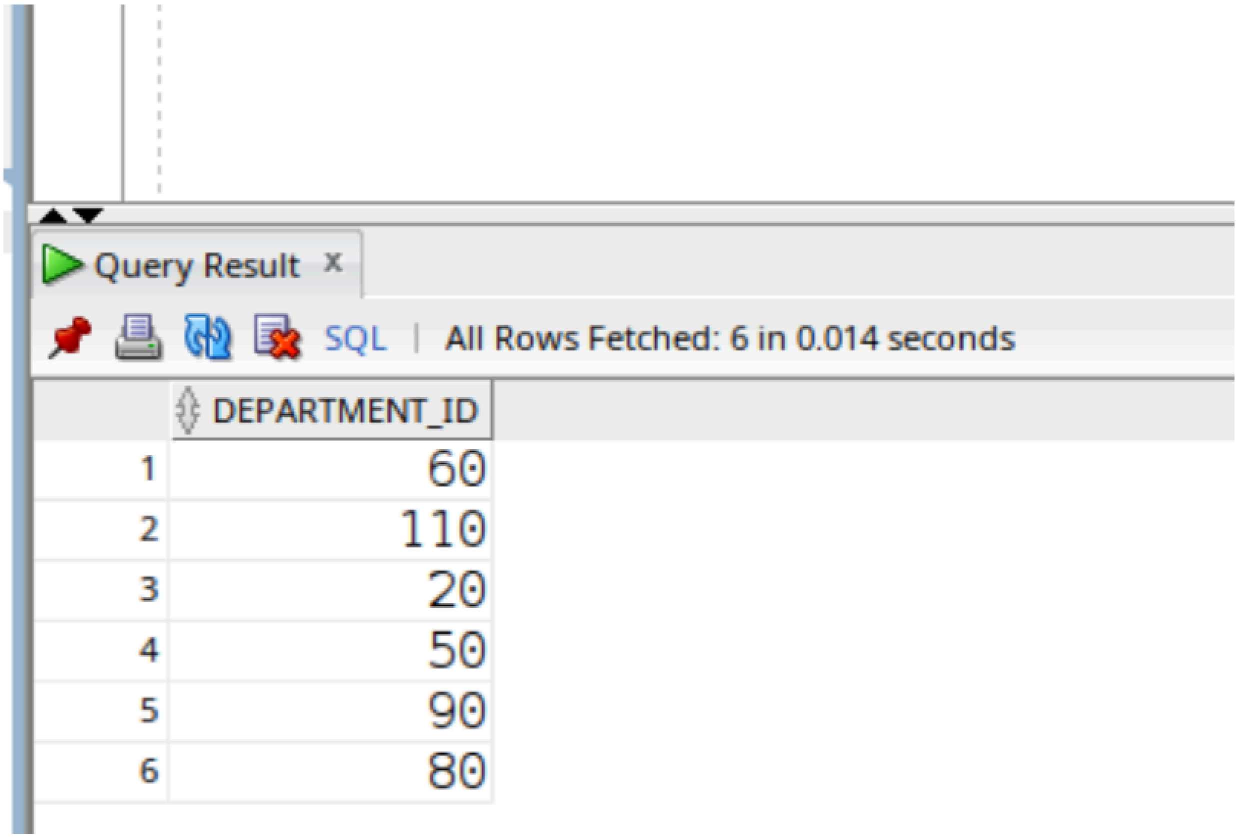
Query Result

SQL | Fetched 50 rows in 0.055 seconds

	JOB_ID
1	AC_ACCOUNT
2	AC_MGR
3	AD_ASST
4	AD_PRES
5	AD_VP
6	AD_VP
7	FI_ACCOUNT
8	FI_ACCOUNT
9	FI_ACCOUNT
10	FI_ACCOUNT
11	FI_ACCOUNT
12	FI_MGR
13	HR_REP
14	IT_PROG
15	IT_PROG
16	IT_PROG
17	IT_PROG
18	IT_PROG

Q17:

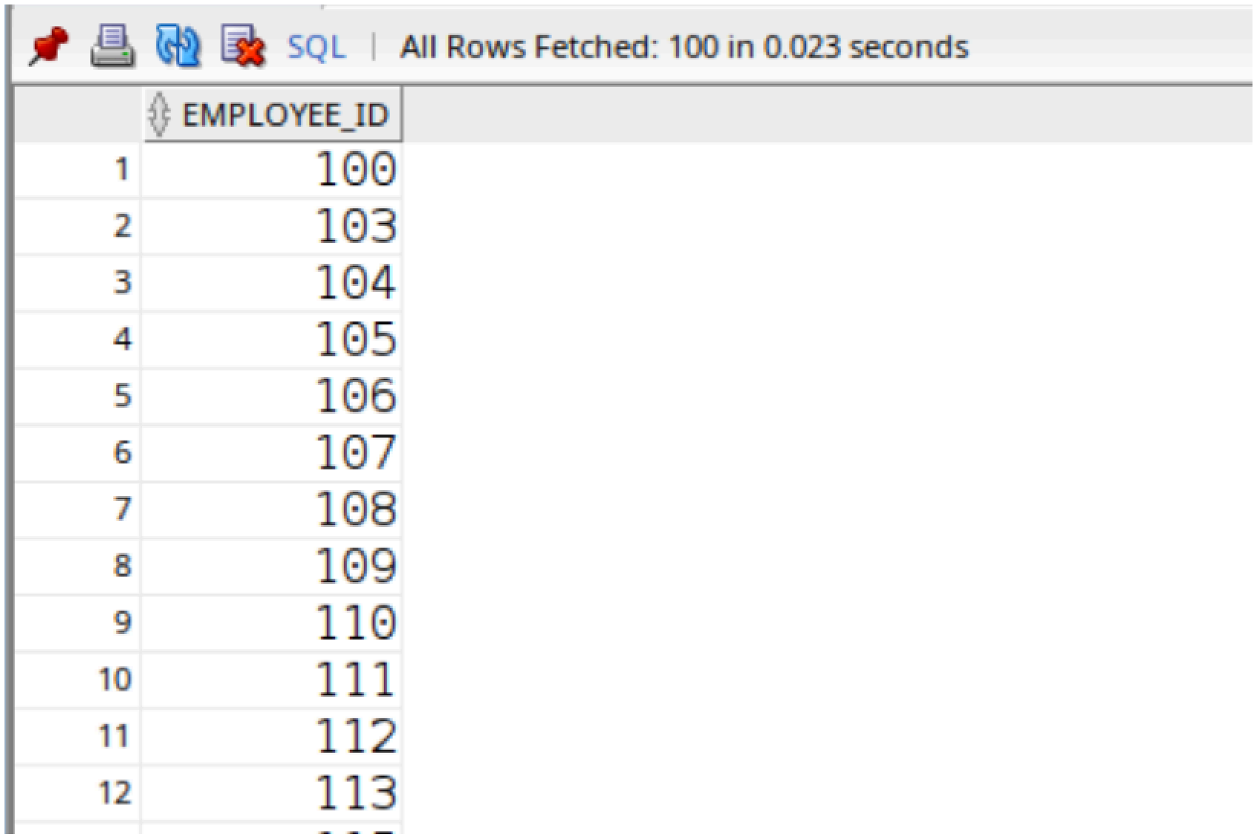
```
select department_id from employees
intersect
select department_id from job_history;
```



	DEPARTMENT_ID
1	60
2	110
3	20
4	50
5	90
6	80

Q18:

```
select employee_id from employees
minus
select e.employee_id from employees e join job_history j
on e.employee_id=j.employee_id where e.job_id <> j.job_id;
```

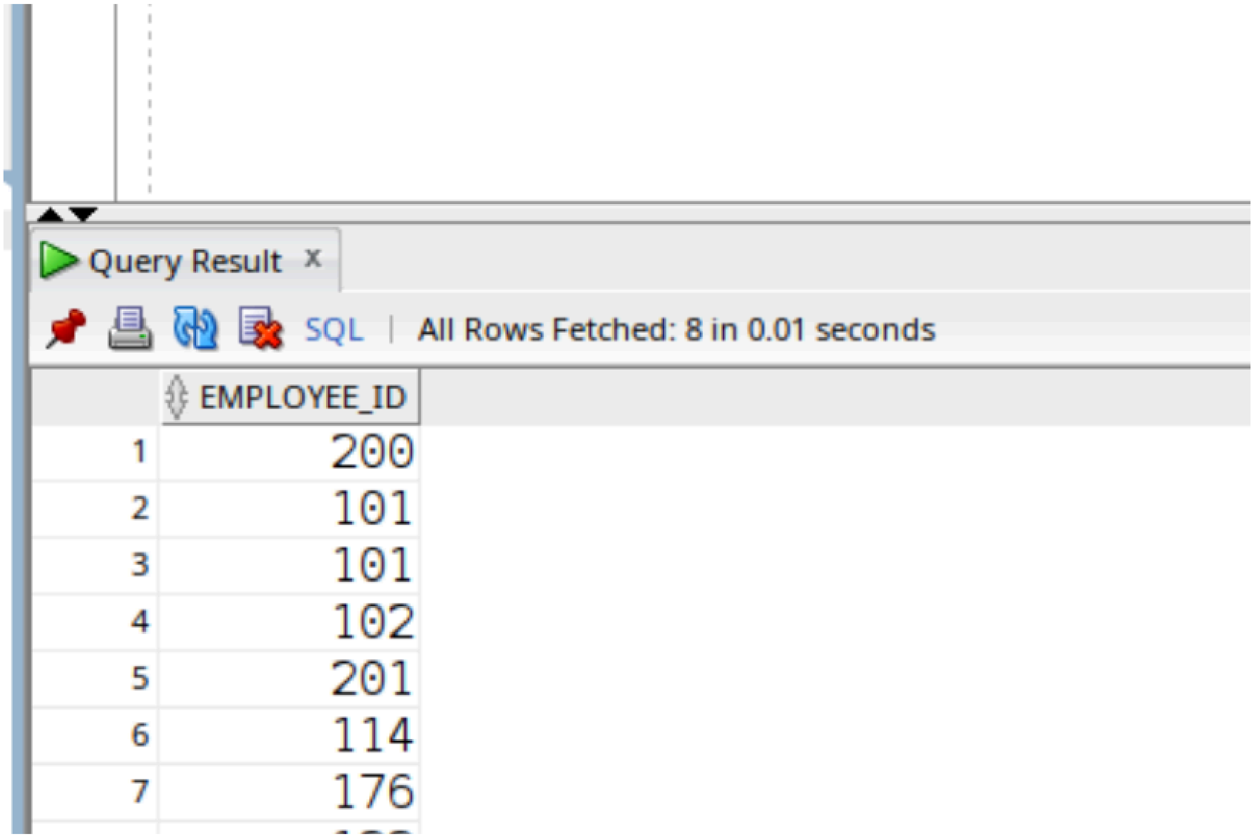


SQL | All Rows Fetched: 100 in 0.023 seconds

	EMPLOYEE_ID
1	100
2	103
3	104
4	105
5	106
6	107
7	108
8	109
9	110
10	111
11	112
12	113

Q19:

```
select e.employee_id from employees e join job_history j
on e.employee_id=j.employee_id where e.job_id <> j.job_id;
```



Query Result ×

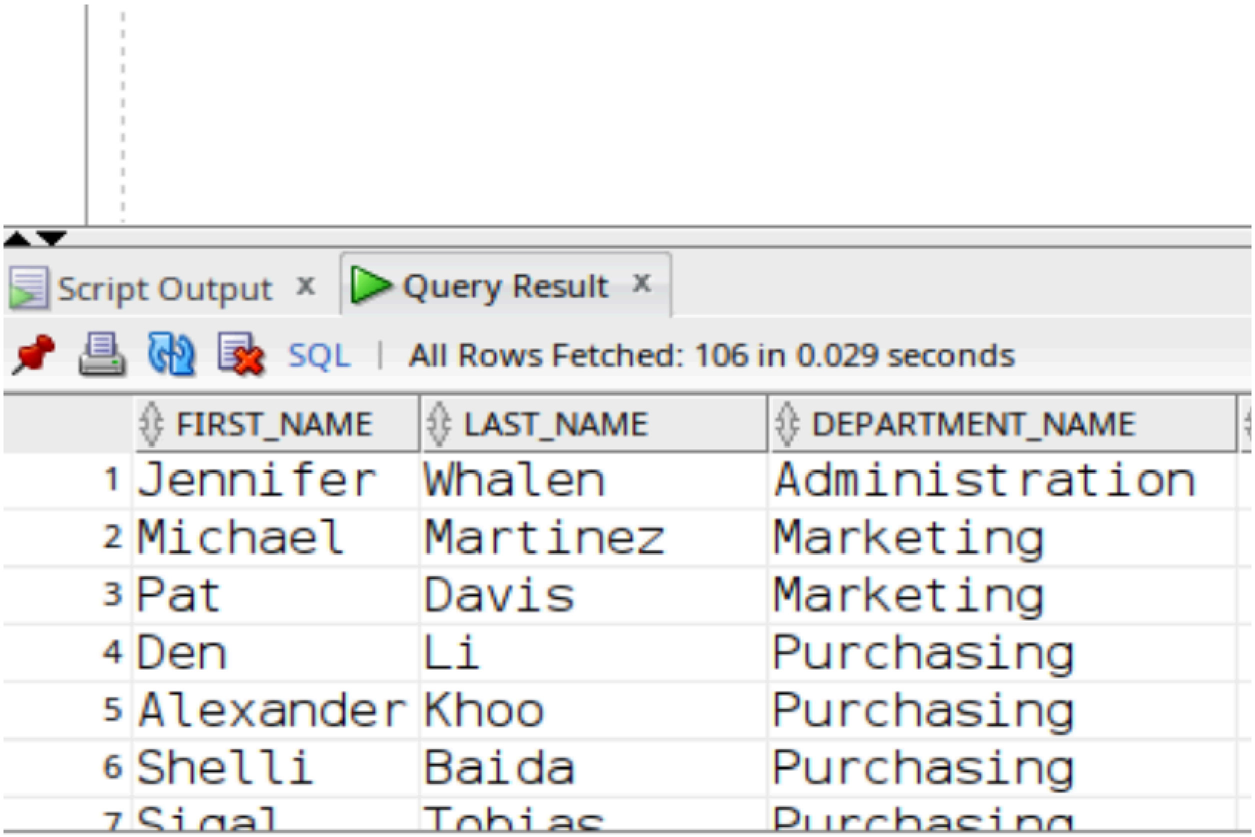
SQL | All Rows Fetched: 8 in 0.01 seconds

	EMPLOYEE_ID
1	200
2	101
3	101
4	102
5	201
6	114
7	176

Q20:

```
create or replace view employee_info as
select first_name,last_name,department_name,salary
from employees join departments
on employees.department_id=departments.department_id;

select * from employee_info;
```



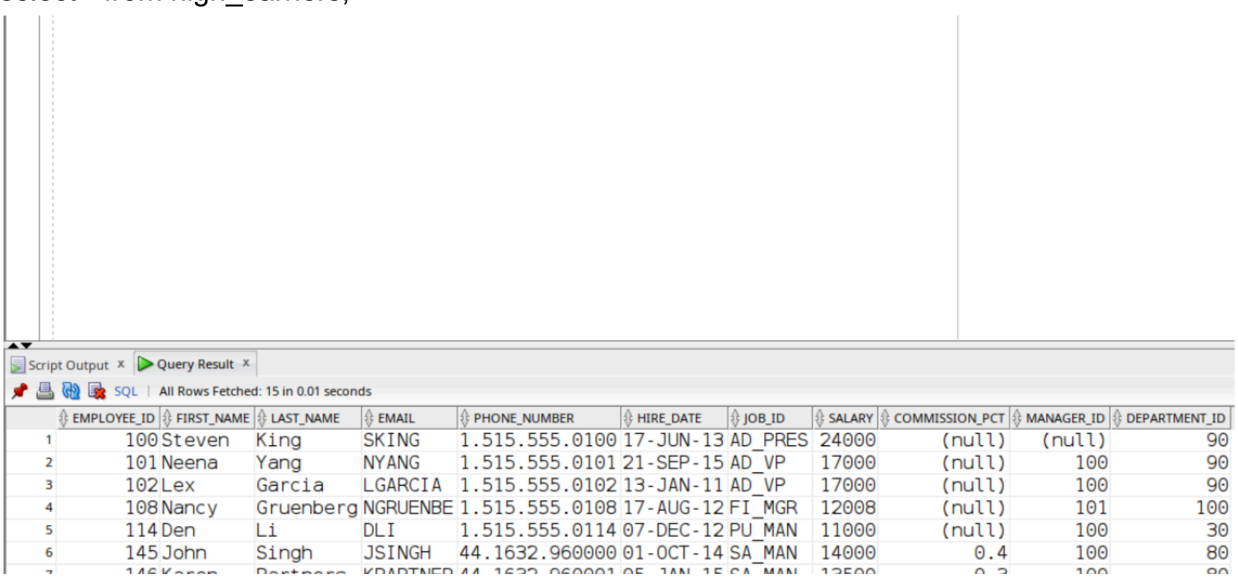
Script Output | Query Result

SQL | All Rows Fetched: 106 in 0.029 seconds

	FIRST_NAME	LAST_NAME	DEPARTMENT_NAME
1	Jennifer	Whalen	Administration
2	Michael	Martinez	Marketing
3	Pat	Davis	Marketing
4	Den	Li	Purchasing
5	Alexander	Khoo	Purchasing
6	Shelli	Baida	Purchasing
7	Sigal	Tobias	Purchasing

Q21:

create or replace view high_earners as
select * from employees where salary>10000;
select * from high_earners;



Script Output | Query Result

SQL | All Rows Fetched: 15 in 0.01 seconds

	EMPLOYEE_ID	FIRST_NAME	LAST_NAME	EMAIL	PHONE_NUMBER	HIRE_DATE	JOB_ID	SALARY	COMMISSION_PCT	MANAGER_ID	DEPARTMENT_ID
1	100	Steven	King	SKING	1.515.555.0100	17-JUN-13	AD_PRES	24000	(null)	(null)	90
2	101	Neena	Yang	NYANG	1.515.555.0101	21-SEP-15	AD_VP	17000	(null)	100	90
3	102	Lex	Garcia	LGARCIA	1.515.555.0102	13-JAN-11	AD_VP	17000	(null)	100	90
4	108	Nancy	Gruenberg	NGRUENBE	1.515.555.0108	17-AUG-12	FI_MGR	12008	(null)	101	100
5	114	Den	Li	DLI	1.515.555.0114	07-DEC-12	PU_MAN	11000	(null)	100	30
6	145	John	Singh	JSINGH	44.1632.960000	01-OCT-14	SA_MAN	14000	0.4	100	80

Q22:

create or replace view report_to as
select e.first_name||' ' || e.last_name as employee,
m.first_name||' '||m.last_name as reporting_to
from employees e join employees m on e.manager_id=m.employee_id;
select * from report_to;

Script Output × | Query Result ×

SQL | Fetched 50 rows in 0.023 seconds

	EMPLOYEE	REPORTING_TO
1	Gerald Cambrault	Steven King
2	Alberto Errazuriz	Steven King
3	Adam Fripp	Steven King
4	Lex Garcia	Steven King
5	Payam Kaufling	Steven King
6	Den Li	Steven King
7	Michael Martinez	Steven King

Q23:

create or replace view avg_department_salary as
select department_id, avg(salary) as avg_salary
from employees group by department_id;
select * from avg_department_salary;

Script Output × | Query Result ×

SQL | All Rows Fetched: 12 in 0.009 seconds

	DEPARTMENT_ID	AVG_SALARY
1	90	19333.3333333333333333333333333333333
2	60	5760
3	100	8601.33333333333333333333333333333333
4	30	4150
5	50	3475.55555555555555555555555555555556
6	80	8955.88235294117647058823529411764706

Q24:

create or replace view all_info as
select e.employee_id,e.first_name,e.last_name,e.salary,
d.department_id,d.department_name,l.location_id,l.city
from employees e join departments d on e.department_id=d.department_id
join locations l on d.location_id=l.location_id;
select * from all_info;

Script Output × | Query Result ×

SQL | Fetched 50 rows in 0.032 seconds

	EMPLOYEE_ID	FIRST_NAME	LAST_NAME	SALARY	DEPARTMENT_ID	DEPARTMENT_NAME	LOCATION_ID	CITY
1	100	Steven	King	24000	90	Executive	1700	Seattle
2	101	Neena	Yang	17000	90	Executive	1700	Seattle
3	102	Lex	Garcia	17000	90	Executive	1700	Seattle
4	103	Alexander	James	9000	60	IT	1400	Southlake
5	104	Bruce	Miller	6000	60	IT	1400	Southlake
6	105	David	Williams	4800	60	IT	1400	Southlake

Q25:

create or replace view salary_info as
select * from employees where salary>5000
with check option;
select * from salary_info;

Script Output × Query Result ×

SQL | Fetched 50 rows in 0.019 seconds

	EMPLOYEE_ID	FIRST_NAME	LAST_NAME	EMAIL	PHONE_NUMBER	HIRE_DATE	JOB_ID	SALARY	COMMISSION_PCT	MANAGER_ID	DEPARTMENT_ID
1	100	Steven	King	SKING	1.515.555.0100	17-JUN-13	AD_PRES	24000	(null)	(null)	90
2	101	Neena	Yang	NYANG	1.515.555.0101	21-SEP-15	AD_VP	17000	(null)	100	90
3	102	Lex	Garcia	LGARCIA	1.515.555.0102	13-JAN-11	AD_VP	17000	(null)	100	90
4	103	Alexander	James	AJAMES	1.590.555.0103	03-JAN-16	IT_PROG	9000	(null)	102	60
5	104	Bruce	Miller	BMILLER	1.590.555.0104	21-MAY-17	IT_PROG	6000	(null)	103	60
6	108	Nancy	Gruenberg	NGRUENBE	1.515.555.0108	17-AUG-12	FI_MGR	12008	(null)	101	100

Q26:

create or replace view job_history_view as
select * from job_history
with read only;
select * from job_history;

Script Output × Query Result ×

SQL | All Rows Fetched: 10 in 0.041 seconds

	EMPLOYEE_ID	START_DATE	END_DATE	JOB_ID	DEPARTMENT_ID
1	102	13-JAN-11	24-JUL-16	IT_PROG	60
2	101	21-SEP-07	27-OCT-11	AC_ACCOUNT	110
3	101	28-OCT-11	15-MAR-15	AC_MGR	110
4	201	17-FEB-14	19-DEC-17	MK_REP	20
5	114	24-MAR-16	31-DEC-17	ST_CLERK	50
6	122	01-JAN-17	31-DEC-17	ST_CLERK	50

Q27:

create or replace view all_job_roles as
select e.employee_id,e.job_id as current_job, j.job_id as previous_job
from employees e join
job_history j on e.employee_id=j.employee_id;

select * from all_job_roles;

Script Output × Query Result ×

SQL | All Rows Fetched: 10 in 0.009 seconds

	EMPLOYEE_ID	CURRENT_JOB	PREVIOUS_JOB
1	200	AD_ASST	AC_ACCOUNT
2	200	AD_ASST	AD_ASST
3	101	AD_VP	AC_ACCOUNT
4	101	AD_VP	AC_MGR
5	102	AD_VP	IT_PROG
6	201	MK_MAN	MK_REP

Q28:

update emp_details_view set salary=60000 where employee_id=103;

Q29:

create or replace view high_earner_department as
select employee_id, department_id, salary from employees e
where salary >(select avg(salary) from employees
where department_id=e.department_id);
select * from high_earner_department;

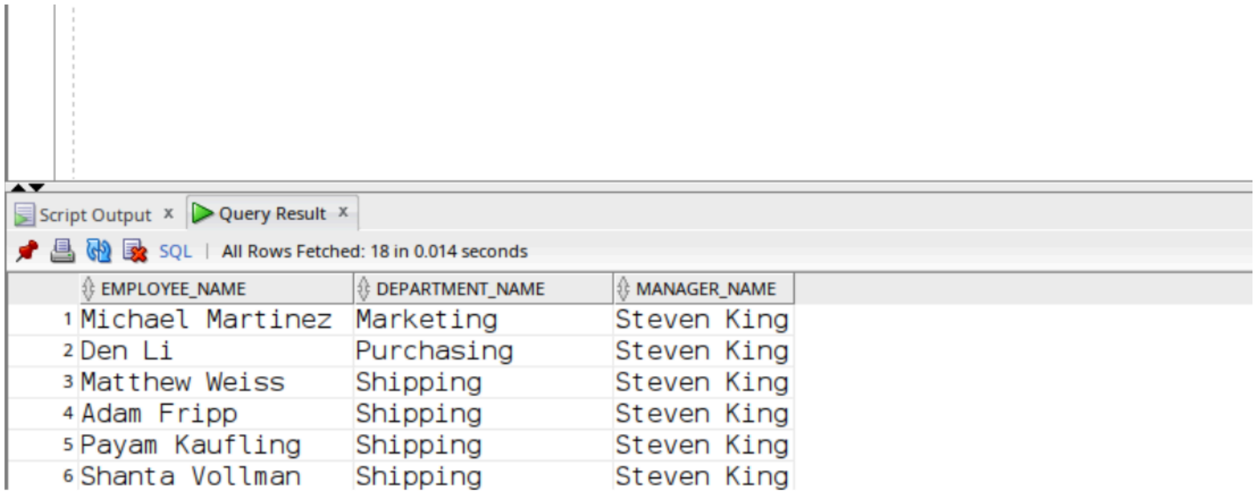
Script Output × Query Result ×

SQL | All Rows Fetched: 38 in 0.041 seconds

	EMPLOYEE_ID	DEPARTMENT_ID	SALARY
1	100	90	24000
2	103	60	9000
3	104	60	6000
4	108	100	12008
5	109	100	9000
6	114	30	11000

Q30:

```
create or replace view collaboration_hiearchy_view as
select e.first_name||' '||e.last_name as employee_name,
d.department_name, m.first_name||' '||m.last_name as manager_name,
from employees e join employees m on e.manager_id=m.employee_id
join departments d on e.department_id=d.department_id where e.department_id <>
m.department_id;
select * from collaboration_hiearchy_view;
```



	EMPLOYEE_NAME	DEPARTMENT_NAME	MANAGER_NAME
1	Michael Martinez	Marketing	Steven King
2	Den Li	Purchasing	Steven King
3	Matthew Weiss	Shipping	Steven King
4	Adam Fripp	Shipping	Steven King
5	Payam Kaufling	Shipping	Steven King
6	Shanta Vollman	Shipping	Steven King

Q31:

```
create or replace view emp_city_view as
select e.employee_id,e.first_name,e.last_name,l.city
from employees e
join departments d on e.department_id=d.department_id
join locations l on d.location_id=l.location_id;

select *from emp_city_view
where city in (
    select city
    from emp_city_view
    group by city
    having count(*)>5
);
```

Script Output × | Query Result ×

SQL | Fetched 50 rows in 0.037 seconds

	EMPLOYEE_ID	FIRST_NAME	LAST_NAME	CITY
1	100	Steven	King	Seattle
2	101	Neena	Yang	Seattle
3	102	Lex	Garcia	Seattle
4	108	Nancy	Gruenberg	Seattle
5	109	Daniel	Faviet	Seattle
6	110	John	Chen	Seattle

Q32:

create or replace view emp_dept_salary as
select e.employee_id,e.first_name,e.last_name,e.salary,d.department_id,
d.department_name,avg(e.salary) over (partition by d.department_id) as avg_dept_salary
from employees e
join departments d on e.department_id=d.department_id;

select * from emp_dept_salary
where salary>avg_dept_salary;

Script Output × | Query Result ×

SQL | All Rows Fetched: 38 in 0.043 seconds

	EMPLOYEE_ID	FIRST_NAME	LAST_NAME	SALARY	DEPARTMENT_ID	DEPARTMENT_NAME	AVG_DEPT_SALARY
1	201	Michael	Martinez	13000	20	Marketing	
2	114	Den	Li	11000	30	Purchasing	
3	120	Matthew	Weiss	8000	50	Shipping	3475.555555555555555555555555555
4	121	Adam	Fripp	8200	50	Shipping	3475.555555555555555555555555555
5	122	Payam	Kaufling	7900	50	Shipping	3475.555555555555555555555555555
6	123	Shanta	Vollman	6500	50	Shipping	3475.555555555555555555555555555

Q33:

create or replace view dept_count_view as
select d.department_id,d.department_name,count(e.employee_id) as total_employees
from departments d
left join employees e on d.department_id=e.department_id

group by d.department_id,d.department_name;

select * from dept_count_view
where total_employees>(select avg(total_employees) from dept_count_view);

Script Output × | Query Result ×

SQL | All Rows Fetched: 5 in 0.017 seconds

	DEPARTMENT_ID	DEPARTMENT_NAME	TOTAL_EMPLOYEES
1	30	Purchasing	6
2	50	Shipping	45
3	60	IT	5
4	80	Sales	34
5	100	Finance	6

Q34:

create or replace view emp_manager_salary_view as
select e.first_name||' '||e.last_name as employee_name,
e.salary as employee_salary,m.first_name||' '||m.last_name as manager_name,
m.salary as manager_salary from employees e
join employees m on e.manager_id=m.employee_id;

select *from emp_manager_salary_view
where employee_salary>manager_salary;

Script Output × | Query Result ×

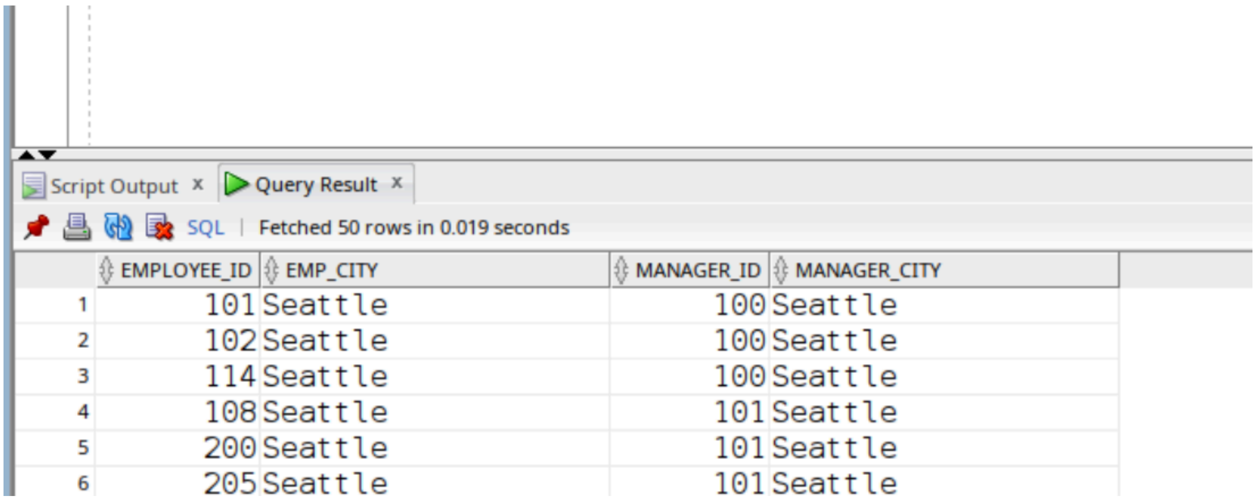
SQL | All Rows Fetched: 2 in 0.006 seconds

	EMPLOYEE_NAME	EMPLOYEE_SALARY	MANAGER_NAME	MANAGER_SALARY
1	Lisa Ozer	11500	Gerald Cambrault	11000
2	Ellen Abel	11000	Eleni Zlotkey	10500

Q35:

```
create or replace view emp_manager_city_view as
select e.employee_id,l1.city as emp_city,m.employee_id as manager_id,
l2.city as manager_city
from employees e
join employees m on e.manager_id=m.employee_id
join departments d1 on e.department_id=d1.department_id
join locations l1 on d1.location_id=l1.location_id
join departments d2 on m.department_id=d2.department_id
join locations l2 on d2.location_id=l2.location_id;

select *from emp_manager_city_view
where emp_city=manager_city;
```

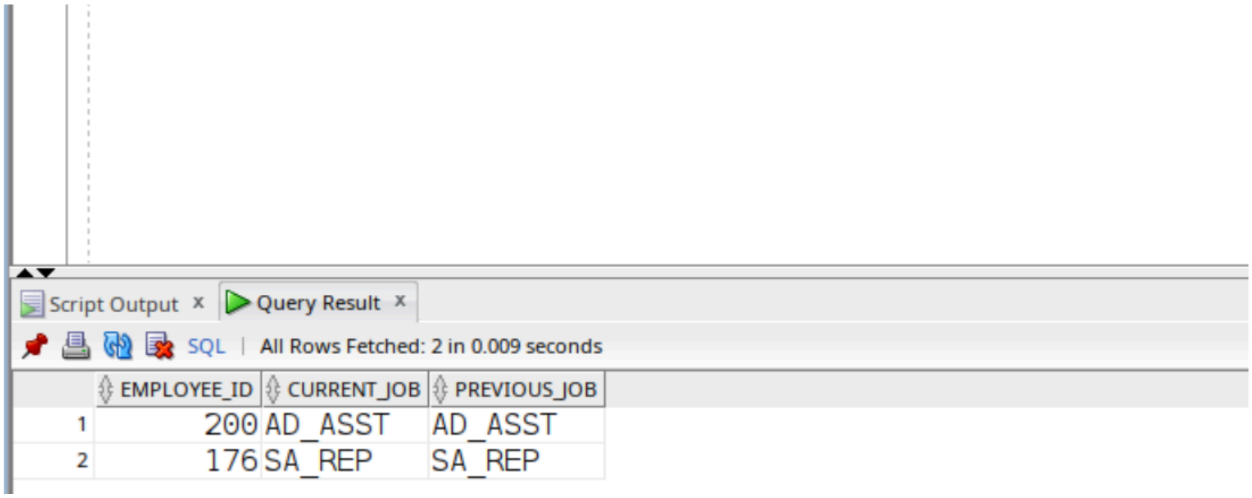


	EMPLOYEE_ID	EMP_CITY	MANAGER_ID	MANAGER_CITY
1	101	Seattle	100	Seattle
2	102	Seattle	100	Seattle
3	114	Seattle	100	Seattle
4	108	Seattle	101	Seattle
5	200	Seattle	101	Seattle
6	205	Seattle	101	Seattle

Q36:

```
create or replace view all_job_roles as
select e.employee_id,e.job_id as current_job,j.job_id as previous_job
from employees e
join job_history j on e.employee_id=j.employee_id;

select * from all_job_roles
where current_job=previous_job;
```

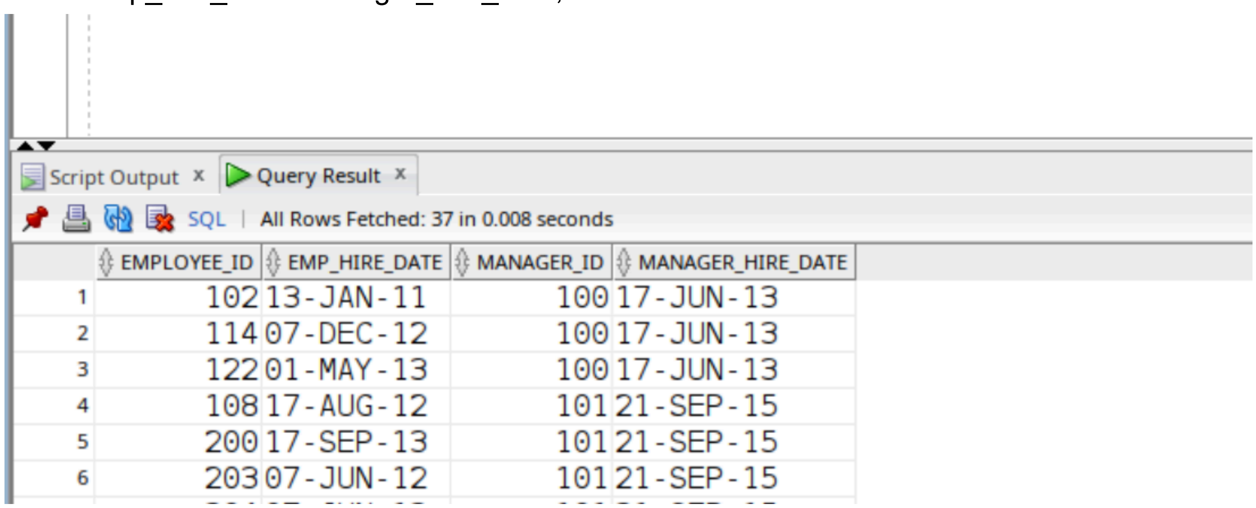


	EMPLOYEE_ID	CURRENT_JOB	PREVIOUS_JOB
1	200	AD_ASST	AD_ASST
2	176	SA_REP	SA_REP

Q37:

create or replace view emp_manager_hire_view as
select e.employee_id,e.hire_date as emp_hire_date,
m.employee_id as manager_id,m.hire_date as manager_hire_date
from employees e
join employees m on e.manager_id=m.employee_id;

select *from emp_manager_hire_view
where emp_hire_date<manager_hire_date;



	EMPLOYEE_ID	EMP_HIRE_DATE	MANAGER_ID	MANAGER_HIRE_DATE
1	102	13-JAN-11	100	17-JUN-13
2	114	07-DEC-12	100	17-JUN-13
3	122	01-MAY-13	100	17-JUN-13
4	108	17-AUG-12	101	21-SEP-15
5	200	17-SEP-13	101	21-SEP-15
6	203	07-JUN-12	101	21-SEP-15

Q38:

create or replace view emp_dept_job_view as
select e.employee_id,e.first_name,e.last_name,e.salary,d.department_id,
d.department_name,j.job_id,j.job_title
from employees e
join departments d on e.department_id=d.department_id

join jobs j on e.job_id=j.job_id;

update emp_dept_job_view
set salary=65000
where employee_id=101;